Partial Differential Equations (PDE)
Assignment 3

[Heat/Diffusion equation]

(a) Verify $\int_{-\infty}^{\infty} e^{-y^2} dy = \sqrt{\pi}$.

(b) Verify

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} e^{-i\lambda x} dx = e^{-i\mu\lambda - \frac{1}{2}\sigma^2\lambda^2}.$$

Recall $\mathcal{N}(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ is Gaussian distribution.

(c) Find the Fourier transformation of $e^{-\eta x^2}$, for any constant $\eta > 0$.

(d) Use the Fourier transformation to show $u(x,t) = (g_t \star \phi)(x)$, where $\star$ denotes convolution and $g_t(x) := \mathcal{N}(x;0,2a^2t)$ is heat kernel, solves

$$\begin{cases} u_t = a^2 u_{xx} \\ u(x,0) = \phi(x) \end{cases}$$